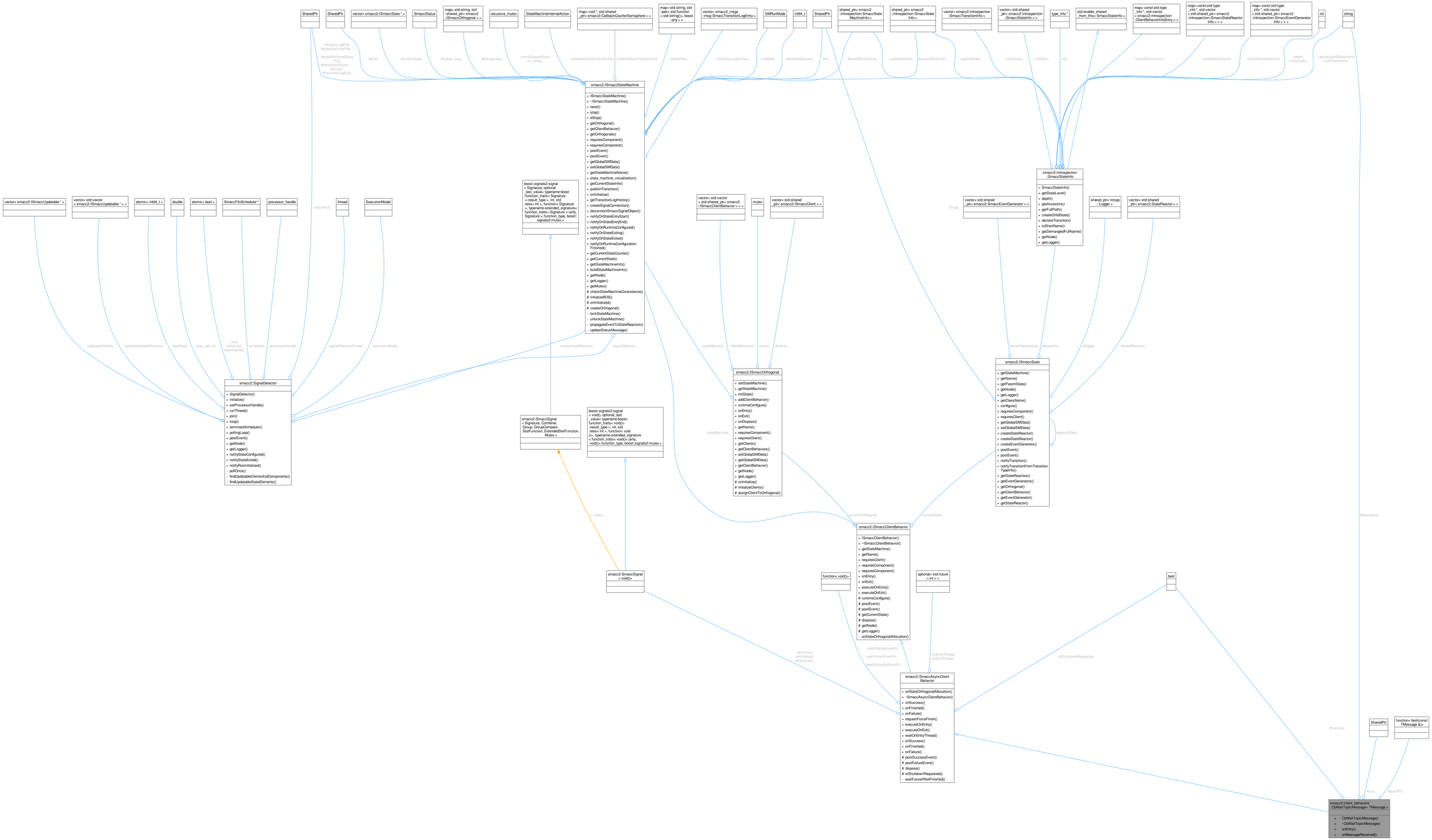
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost:: :any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
shared_ptr< smacc2 ::introspection::SmaccState Info >
vector< smacc2::introspection ::ISmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::ISmaccStateInfo > >
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
int
string
#eventsLogPub_ #stateMachinePub_
#stateMachineStatus Pub_ #transitionHistory Service_ #transitionLogPub_
#timer_
#currentState_
#status_msg_
#orthogonals_
-eventQueueMutex_ -m_mutex_
+stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
+runMode_
-stateSeqCounter_
#sm_
#stateMachineInfo_
+stateMachine_
#currentStateInfo_
+parentState_
+transitions_
+children_
+id_
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+depth_ +stateIndex_
+demangledStateName +fullStateName
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
atomic< bool >
SmaccFifoScheduler *
processor_handle
-updatableClients_
-updatableStateElements_
-lastState_
-loop_rate_hz
-end_ -initialized_ -isInitialized_
-scheduler_
-processorHandle_
-statusPub_
thread
ExecutionModel
-signalDetectorThread_
-executionModel_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
-smaccStateMachine_
-signalDetector_
vector< std::vector < smacc2::ISmaccClientBehavior > >
mutex
vector< std::shared _ptr< smacc2::ISmaccClient > >
-stateMachine_
-clientBehaviors_
-mutex_
#clients_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
#mode_
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
-eventGenerators_
-stateInfo_
-logger_
#stateReactors_
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::SmaccSignal < void() >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
-stateMachine_
< void()>
-currentOrthogonal
-currentState
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
function< void()>
optional< std::future < int > >
-onFailure_ -onFinished_ -onSuccess_
-postFailureEventFn_
-postFinishEventFn_
-postSuccessEventFn_
-onEntryThread_ -onExitThread_
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
bool
-isShutdownRequested_
#topicname_
#success
SharedPtr
function< bool(const TMessage &)>
#sub_
#guardFn_
smacc2::client_behaviors ::CbWaitTopicMessage< TMessage >
+ CbWaitTopicMessage()
+ ~CbWaitTopicMessage()
+ onEntry()
+ onMessageReceived()